TECHCRUSH AI/ML BOOTCAMP — CAPSTONE PROJECT

Machine Learning and Deep Learning for Automated Water Quality Detection and Potability Classification

Cohort: TechCrush Cohort 8 | **Team:** Group 10 | **Facilitators:** Micheal, Lamzey, and Ojuri | **Date:** October 2026

Team Contribution & Pipeline Ownership Matrix

Team Member	Project Role	Core Pipeline Responsibilities
Deborah	Data Acquisition & Cleaning	Dataset provenance, metadata documentation, missing value audit, and median imputation rationale.
Surafel Esayas	Exploratory Data Analysis (EDA)	Univariate distributions, 61/39 class imbalance analysis, correlation heatmap, and observations.
Araad HOF	Preprocessing & Modelling	WHO safe flags engineering, 80/20 stratified split, StandardScaler, Random Forest & Gradient Boosting.
Ostino (Lead)	Deep Learning, Evaluation & Compilation	Neural Network (ANN) architecture, evaluation comparison, confusion matrices, and final report compilation.

Abstract

Access to safe drinking water is a fundamental prerequisite for human health, economic productivity, and ecological sustainability. Traditional laboratory-based water quality testing is slow, expensive, and unable to provide immediate early warnings during rapid contamination shocks. This capstone project designs and evaluates an automated water potability classification pipeline using classical machine learning and deep learning. Utilizing the benchmarked Kaggle Water Potability dataset (3,276 samples across 9 physicochemical attributes), we executed rigorous data cleaning via class-conditional median imputation, exploratory data analysis, domain-driven feature engineering, and stratified data partitioning. We trained and evaluated three distinct classification paradigms: Random Forest (Bagging), Gradient Boosting (Boosting), and a Multi-Layer Perceptron (Artificial Neural Network) regularized with Batch Normalization and Dropout. Experimental results indicate that Tree Ensemble architectures strongly outperform the deep neural network on this tabular task. Gradient Boosting attained the highest overall performance with an Accuracy of 80.49%, a Precision of 82.99% for potable water, and an ROC-AUC of 0.8912. Furthermore, we evaluate the asymmetric public health implications of classification errors, establishing why prioritizing the minimization of false positives is vital for operational municipal water distribution safety.

1. Introduction

Access to clean drinking water is recognized by the United Nations as a fundamental human right under Sustainable Development Goal 6 (Clean Water and Sanitation). Despite global sanitation initiatives, the World Health Organization (WHO, 2017) estimates that over two billion people rely on drinking water sources contaminated with microbial or chemical hazards. Waterborne diseasesâ€"such as cholera, dysentery, typhoid fever, and hepatitis Aâ€"cause hundreds of thousands of preventable deaths annually. Beyond acute pathogens, chronic ingestion of chemical contaminants, abnormal pH levels, or disinfection by-products induces chronic renal damage, neurological impairment, and elevated cancer risk.

In conventional water management, quality assurance relies on periodic physical sample collection followed by laboratory benchtop chemical analysis (including titration, chromatography, and spectrometry). While precise, this centralized approach presents major limitations:

1. **Turnaround Latency:** Transporting samples from remote reservoirs to testing facilities often takes days or weeks, during which communities may consume hazardous water.

2. **Economic Constraints:** High-precision laboratory analysis demands expensive equipment, chemical reagents, and trained personnel, making frequent municipal testing cost-prohibitive.

3. **Absence of Early Warnings:** Periodic manual testing cannot capture sudden contamination shocks caused by flash floods, agricultural runoff, pipe bursts, or industrial chemical leaks.

Advances in multiparameter sensor telemetry and machine learning present an opportunity to automate water potability screening. By capturing continuous physicochemical signals (such as pH, dissolved solids, conductivity, and turbidity), predictive models can synthesize multivariate data streams to instantaneously classify water as potable (safe) or non-potable (unsafe).

This capstone project develops, evaluates, and compares multiple machine learning and deep learning architectures to predict water potability. In alignment with the TechCrush AI/ML Bootcamp grading criteria, this report justifies every engineering decisionâ€"from median imputation and feature scaling to algorithm selection, hyperparameter tuning, and error cost analysisâ€"providing a scientifically grounded framework for real-world environmental deployment.

2. Dataset Acquisition and Description

2.1 Provenance and Sourcing

The empirical dataset utilized in this project was obtained from the benchmarked **Kaggle Water Quality / Water Potability Dataset** curated by Aditya Kadiwal (2020). The data aggregates measurements gathered from natural water bodies, municipal reservoirs, and water treatment facilities, providing a standard tabular benchmark for binary potability classification.

2.2 Dataset Attributes and WHO Permissible Limits

The dataset contains **3,276 individual observations** and **10 attributes**: 9 continuous numerical physicochemical features and 1 binary ground-truth target (`Potability`).

1. **`ph` (pH Value):** Logarithmic scale measuring hydrogen ion concentration. The WHO recommends an acceptable drinking water range between **6.5 and 8.5**. Values below 6.5 corrode plumbing infrastructure and leach toxic metals, while values above 8.5 cause mineral encrustation and reduce chlorine disinfection potency.

2. **`Hardness` (mg/L):** Measures dissolved polyvalent metallic ions, primarily calcium (Ca^{2+}) and magnesium (Mg^{2+}). Levels exceeding 180 mg/L indicate very hard water that causes pipeline scaling.

3. **`Solids` (Total Dissolved Solids - TDS in ppm):** Represents dissolved organic and inorganic minerals and salts. Water exceeding 1,000 ppm exhibits an adverse mineral taste and induces digestive discomfort.

4. **`Chloramines` (ppm):** Formed by combining chlorine and ammonia for secondary disinfection across municipal pipes. Permissible safe levels range up to **4.0 ppm**.

5. **`Sulfate` (mg/L):** Naturally occurring minerals derived from geological deposits. Concentrations exceeding **250 to 500 mg/L** impart a bitter taste and cause cathartic (laxative) effects.

6. **`Conductivity` (μS/cm):** Electrical conductance reflecting ionic concentration. Potable water typically registers below **400 μS/cm**.

7. **`Organic_carbon` (TOC in ppm):** Quantifies decaying organic matter. High levels (> 2 to 4 ppm) encourage bacterial regrowth and react with chlorine to form carcinogenic disinfection by-products.

8. **`Trihalomethanes` (THMs in μg/L):** Volatile by-products formed when chlorine reacts with dissolved organic carbon. Chronic consumption is linked to cancer; the WHO safe limit is **$\le 80\,\mu$g/L**.

9. **`Turbidity` (NTU):** Measures cloudiness caused by suspended microscopic particles. The WHO specifies drinking water should remain below **5 NTU** (ideally < 1 NTU for disinfection efficacy).

10. **`Potability` (Target):** Binary indicator where `1` designates safe/potable water, and `0` designates unsafe/non-potable water.

2.3 Known Dataset Limitations

To contextualize our findings, we acknowledge two inherent limitations:

- **Absence of Pathogen Assays:** The dataset lacks microbiological colony counts (e.g., *E. coli* or coliform bacteria). Water meeting all 9 chemical thresholds could still be hazardous if biological pathogens are present.

- **Aggregated Spatiotemporal Context:** The dataset lacks geographic coordinates, timestamps, or seasonal tags, precluding seasonal runoff modeling.

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3. Data Cleaning and Outlier Analysis

Sensor-based environmental records frequently contain unrecorded measurements, transmission dropouts, and hydrological shocks. We conducted a structured cleaning audit to prepare the dataset.

3.1 Duplicate and Missing Value Audit

A vectorized duplicate check across all 3,276 records revealed **0 duplicate rows**. However, a missingness audit identified unrecorded values across three key chemical features:

- **`Sulfate`:** 781 missing values (**23.84%** of the dataset).

- **`ph`:** 491 missing values (**14.99%** of the dataset).

- **`Trihalomethanes`:** 162 missing values (**4.95%** of the dataset).

The remaining six features and the target variable contained zero missing values.

3.2 Cleaning Decisions and Methodological Justifications

Rejection of Row Deletion (`dropna`)

A standard initial impulse is complete-case row deletion (`df.dropna()`). However, dropping all rows with missing values would discard **1,265 out of 3,276 samples**, destroying **38.61% of our entire dataset**. Deleting nearly 40% of observations drastically shrinks training sample size, removes critical boundary samples, and introduces sampling bias (since sensor failures frequently correlate with extreme weather). Complete-case deletion was therefore rejected.

Selection of Class-Conditional Median Imputation

We evaluated mean, KNN, and median imputation, selecting **Median Imputation conditioned on the target class (`Potability`)**:

1. **Robustness Against Distribution Skewness:** Features like `Sulfate` and `ph` exhibit non-normal distributions with notable tails. While the arithmetic mean is sensitive to extreme values, the median

provides a robust measure of central tendency.

2. **Preserving Subtle Class Distributions:** Imputing global medians across the whole dataset flattens statistical separation between classes. Calculating medians separately for non-potable (`Potability = 0`) and potable (`Potability = 1`) samples preserves class-specific distributional characteristics without introducing cross-class contamination.

3.3 Outlier Handling and Environmental Grounding

Using the Interquartile Range (IQR) method ($1.5 \times \text{IQR}$), extreme observations were detected in `Solids` (reaching 60,000 ppm), `Conductivity`, and `Turbidity`. Rather than deleting these observations as errors, we referenced hydrological principles. Flash floods, agricultural runoff, and mineral leaching routinely cause sharp, legitimate surges in total dissolved solids and turbidity. Because real-world monitoring systems must operate under these exact conditions, deleting them would produce an unrealistically sanitized dataset. We retained these observations and addressed scale leverage during preprocessing through feature scaling.

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4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to inspect class distributions, feature spread, and multi-variable correlations, connecting every visual insight directly to downstream modeling choices.

4.1 Target Class Imbalance

Analysis of `Potability` revealed the following distribution:

- **Class 0 (Non-potable / Unsafe):** 1,998 samples (**60.99%**).

- **Class 1 (Potable / Safe to Drink):** 1,278 samples (**39.01%**).

Observation and Impact on Modeling:

The dataset exhibits a moderate class imbalance (approx. 61% to 39%).

Key Takeaway: A naive baseline model predicting only the majority class (non-potable) would achieve an apparent accuracy of 60.99% while failing to identify any safe water. Therefore, **Accuracy alone is misleading**. We must prioritize **Precision, Recall, and F1-Score for Class 1**, and enforce **Stratified Splitting** across all validation sets.

4.2 Feature Distributions and Overlap

Kernel Density Estimation (KDE) and probability density histograms were plotted for all 9 continuous variables grouped by potability class:

- Features such as `Hardness`, `Chloramines`, `Sulfate`, `Conductivity`, and `Turbidity` conform closely to symmetric, bell-shaped Gaussian curves.

- Crucially, the distributions for potable (Class 1) and non-potable (Class 0) water overlap substantially across every single feature dimension (e.g., mean pH is 7.07 for potable water vs. 7.08 for non-potable water).

Observation and Impact on Modeling:

No single chemical feature acts as an isolated linear separator. Safe water cannot be determined by a single threshold on any one feature. Instead, potability is governed by complex, multi-dimensional boundary interactions. This observation rules out simple linear classifiers and justifies using non-linear ensemble models and neural networks.

4.3 Correlation Matrix Analysis

A Pearson correlation heatmap was generated across all features and the target variable:

- Linear correlation coefficients between the 9 features and `Potability` were uniformly low, ranging from $|r| = 0.003$ to $|r| = 0.038$.

- Pairwise correlations between features were also near zero ($|r| < 0.08$).

Observation and Impact on Modeling:

The absence of linear correlation confirms that linear models (such as basic Logistic Regression) cannot easily separate the classes. Furthermore, the lack of multicollinearity confirms that each chemical sensor provides independent information, meaning all 9 features should be retained without aggressive dimensionality reduction.

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5. Data Preprocessing and Feature Engineering

Data preprocessing converted the cleaned tabular measurements into an optimized numerical format suitable for tree ensembles and neural network backpropagation.

5.1 Domain-Driven Feature Engineering

To inject domain-informed signals derived from World Health Organization guidelines, we engineered three new biochemical features:

1. **`ph_in_safe_range` (Binary):**

$$1 \text{ if } 6.5 \le \text{ph} \le 8.5 \text{ else } 0$$

Rationale: Explicitly identifies samples within the WHO safe window, helping models capture non-linear boundary limits without searching for dual split thresholds.

2. **`turbidity_safe` (Binary):**

$$1 \text{ if Turbidity} \le 5.0 \text{ else } 0$$

Rationale: Directly marks compliance with the WHO maximum turbidity threshold for effective disinfection.

3. **`disinfectant_mineral_ratio` (Continuous):**

$$\frac{\text{Chloramines}}{\text{Sulfate} + 10^{-5}}$$

Rationale: Measures the relative proportion of active chemical disinfectants to background mineral concentration.

These engineered features expanded our feature space from 9 raw parameters to **12 total features**.

5.2 Stratified Train-Test Splitting

The data was split into a **Training Set (80% / 2,620 samples)** and an unseen **Test Set (20% / 656 samples)** using a fixed seed (`random_state=42`).

Justification for Stratification:

We enforced `stratify=y` to preserve the 61/39 class ratio across both partitions. Without stratification, random sampling risks uneven distribution of the minority potable class, creating distribution drift between training and testing.

5.3 Feature Scaling

We applied z-score standardization using Scikit-Learnâ€™s `StandardScaler`:

$$z = \frac{x - \mu}{\sigma}$$

Justification for Scaling and Preventing Data Leakage:

Numerical features vary widely in magnitude (`Solids` reaches up to 60,000 ppm, whereas `ph` ranges between 0 and 14). Unscaled features cause severe gradient oscillations during neural network backpropagation. The scaler was **fitted strictly on the training set (`X_train`)** and then applied to transform the test set (`X_test`), strictly preventing test-set data leakage.

6. Modelling

To satisfy the rubric requirement for multiple modeling approaches, we trained, tuned, and evaluated three distinct classification paradigms:

1. **Random Forest Classifier** (Bagging Ensemble)
2. **Gradient Boosting Classifier** (Boosting Ensemble)
3. **Artificial Neural Network (Multilayer Perceptron - ANN)** (Deep Learning)

6.1 Model 1: Random Forest Classifier (Bagging)

- **Theoretical Rationale:** Random Forest (Breiman, 2001) builds an ensemble of de-correlated decision trees trained on bootstrap samples of the data, combining predictions by majority voting. It handles non-linear interactions, resists overfitting through variance reduction, and accommodates tabular outliers effectively.

- **Hyperparameter Configuration:**

- `n_estimators = 200`: Provides sufficient ensemble diversity for variance stabilization.

- `max_depth = 12`: Constrains individual tree depth to avoid memorizing sample noise.

- `min_samples_split = 4`: Requires adequate sample support before splitting internal nodes.

- `class_weight = 'balanced'`: Adjusts sample weights inversely proportional to class frequencies to penalize errors on minority potable samples.

6.2 Model 2: Gradient Boosting Classifier (Boosting)

- **Theoretical Rationale:** In contrast to parallel bagging, Gradient Boosting (Friedman, 2001) builds trees sequentially. Each subsequent tree fits the negative gradient of the loss function (the pseudo-residuals) from preceding estimators, focusing capacity on difficult boundary regions.

- **Hyperparameter Configuration:**

- `n_estimators = 150`: Number of sequential boosting stages.

- `learning_rate = 0.05`: Shrinkage parameter moderating individual tree contributions to prevent premature convergence.

- `max_depth = 4`: Utilizes shallow base learners to maintain low variance.

- `subsample = 0.8`: Stochastic gradient boosting using 80% of data per iteration to reduce inter-tree correlation.

6.3 Model 3: Artificial Neural Network (Multilayer Perceptron - ANN)

- **Theoretical Rationale:** Deep Artificial Neural Networks learn non-linear representations through layers of affine transformations and non-linear activations. We evaluated whether a regularized deep neural network could match or exceed tree ensembles on this task.

- **Architecture and Specifications:**

- **Input Layer:** 12 nodes (standardized inputs).

- **Hidden Layer 1:** Fully connected `Dense` layer with **64 neurons**, `BatchNormalization()`, and `Dropout(0.3)` to disrupt co-adaptation of features.

- **Hidden Layer 2:** Fully connected `Dense` layer with **32 neurons**, `BatchNormalization()`, and `Dropout(0.2)`.

- **Hidden Layer 3:** Fully connected `Dense` layer with **16 neurons**.

- **Output Layer:** Single `Dense` node with `Sigmoid` activation: $\sigma(z) = \frac{1}{1 + e^{-z}}$.

- **Mathematical Justifications for Components:**

- *ReLU in Hidden Layers:* The Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$) was selected over Sigmoid/Tanh because it mitigates vanishing gradients and enables fast backpropagation.

- *Sigmoid in Output Layer:* Essential for binary classification, mapping outputs strictly to $[0.0, 1.0]$ to represent the posterior probability of potability: $P(\text{Potability} = 1 \mid X)$.

- *Loss Function (`binary_crossentropy`):* Standard cross-entropy loss for binary classification:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]$$

- *Optimizer (`Adam`):* Adaptive moment estimation with learning rate $\alpha = 0.001$.

- *Early Stopping:* Monitored `val_loss` with `patience=12` and `restore_best_weights=True` to halt training when validation error plateaued.

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7. Results and Evaluation

All models were evaluated on the unseen test set (656 samples) across standard classification metrics: **Accuracy, Precision (Class 1), Recall (Class 1), F1-Score (Class 1), and ROC-AUC**.

7.1 Model Comparison Summary Table

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================================================================================================================
MODEL EVALUATION SUMMARY BENCHMARK (TEST SET: N = 656)
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Model Architecture         Accuracy     Precision (Class 1)  Recall (Class 1)  F1-Score (Class 1)  ROC-AUC
----------------------------------------------------------------------------------------------------------------
Gradient Boosting          80.49%       82.99%               62.89%            71.56%              0.8912
Random Forest              80.34%       80.09%               66.02%            72.38%              0.8820
Neural Network (ANN)       65.09%       58.94%               34.77%            43.73%              0.6433
================================================================================================================
```

7.2 Analysis of Model Performance

Strong Performance of Tree Ensembles

The experimental results demonstrate that **Ensemble Decision Tree algorithms clearly outperform the Deep Neural Network across all metrics**:

- **Gradient Boosting** achieved the highest overall Accuracy (**80.49%**), the highest Precision for potable water (**82.99%**), and the highest overall discrimination ability with an ROC-AUC of **0.8912**.

- **Random Forest** achieved comparable accuracy (**80.34%**), an ROC-AUC of **0.8820**, and the highest overall Recall for safe water (**66.02%**), yielding the highest balanced F1-Score (**72.38%**).

- Both tree ensembles surpassed the 61% majority-class baseline by nearly 20 percentage points, confirming their capacity to learn non-linear chemical interactions.

Neural Network Underperformance Analysis

The Multi-Layer Perceptron (ANN) achieved **65.09% accuracy**, **58.94% precision**, **34.77% recall**, and an ROC-AUC of **0.6433**. While the ANN learned a non-trivial boundary above the 61% baseline, it struggled with high false negative rates, missing nearly two-thirds of truly potable samples.

Why did the Neural Network underperform?

This result aligns with findings in machine learning research (e.g., Grinsztajn et al., 2022). Deep neural networks rely on smooth continuous manifold projections, which require large dataset volumes to generalize well. On smaller tabular datasets (~3,000 samples) with missing values and overlapping distributions, decision tree algorithms form orthogonal step-function axis splits that mirror physical boundary cutoffs far more effectively than backpropagation-trained hyperplanes.

7.3 Confusion Matrix Analysis

Examining the confusion matrices across the 656 test samples highlights practical operational differences:

- **Gradient Boosting:** Correctly classified 367 out of 400 non-potable samples (91.75% True Negative Rate) and 161 out of 256 potable samples (62.89% True Positive Rate), producing only 33 False Positives.

- **Random Forest:** Correctly classified 358 out of 400 non-potable samples (89.50% True Negative Rate) and 169 out of 256 potable samples (66.02% True Positive Rate), producing 42 False Positives.

- **Neural Network:** Misclassified 167 potable samples as non-potable, demonstrating overly conservative predictions.

7.4 Public Health Domain Interpretation: Asymmetric Cost of Errors

> *[!IMPORTANT]*
>
> ***Domain Risk Analysis:***
>
> *In environmental health, classification errors have asymmetric consequences:*
>
> *- **False Negative (Predicting Potable water as Unsafe):** Leads to unnecessary boiling, chemical re-treatment, or disposal of clean water. This causes economic and operational inconvenience, but presents no toxic threat to human health.*
>
> *- **False Positive (Predicting Contaminated water as Potable):** Toxic water containing excessive trihalomethanes, high sulfates, or abnormal pH is certified for human consumption. Ingesting this water directly causes acute gastrointestinal disease outbreaks, chronic poisoning, and hospitalization.*

From a public health perspective, **minimizing False Positives is paramount**. In practical deployment, **Gradient Boosting is the preferred model**, as its high Precision of **82.99%** ensures that water certified as safe has the lowest rate of false safety clearances. In an operational water treatment station, decision probability thresholds should be set conservatively (e.g., 0.65 or 0.70) to demand high certainty before clearing water for distribution.

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8. Challenges and Lessons Learned

Completing this capstone project provided valuable practical lessons in applied data science and engineering:

1. **Tabular Data and Model Selection:** An initial expectation was that a deep neural network with Batch Normalization and Dropout would outperform classical machine learning models. Seeing Gradient Boosting and Random Forest outperform the neural network by over 15% in accuracy and 0.25 in ROC-AUC was an important empirical takeaway. It demonstrated that algorithm choices should be guided by data structure and sample size, rather than defaulting to architectural complexity.

2. **Feature Overlap and Realistic Limits:** Discovering that distributions of potable and non-potable samples overlap heavily across all 9 features was a major diagnostic challenge. In real-world data science, models cannot create separation where physical boundaries are noisy. This reinforced the value of domain-informed feature engineering (WHO threshold flags) to provide explicit boundary signals.

3. **Imputation Methodology:** Avoiding the temptation to run `df.dropna()` was a vital decision. Recognizing that dropping rows would discard nearly 40% of the dataset prompted us to research and apply class-conditional median imputation, preserving our training sample size and statistical diversity.

4. **What We Would Do Differently:** With additional resources, we would explore synthetic minority oversampling (SMOTE) to improve neural network recall, incorporate microbiological assay telemetry (e.g., coliform counts), and conduct automated hyperparameter searches using Bayesian optimization (Optuna).

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9. Conclusion and Future Work

This capstone project designed and evaluated an automated water potability classification pipeline adhering to the technical and ethical requirements of the TechCrush AI/ML Bootcamp.

Summary of Achievements

- Processed and analyzed 3,276 water quality records across 9 physicochemical attributes.

- Addressed potential 38.6% data loss by applying and defending class-conditional median imputation.

- Executed exploratory data analysis, identifying class imbalance (61/39) and substantial feature overlap, directly informing non-linear model selection.

- Created domain-derived WHO compliance features and applied leakage-free stratified scaling.

- Implemented and evaluated Random Forest, Gradient Boosting, and an Artificial Neural Network.

- Demonstrated that Gradient Boosting achieved the strongest overall performance (**80.49% Accuracy, 82.99% Precision, 0.8912 ROC-AUC**), effectively addressing the public health priority of minimizing false positive safety clearances.

Future Work

Future work will focus on edge-device deployment. By exporting the trained Gradient Boosting model into a lightweight ONNX runtime, the pipeline can run on low-power microcontrollers (such as an ESP32 or Raspberry Pi) interfaced with physical water sensors. Such edge stations could provide autonomous, real-time water safety alerts in rural communities and disaster response operations.

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10. References (APA Format)

1. **Abadi, M., et al.** (2016). TensorFlow: A system for large-scale machine learning. *In 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 16)* (pp. 265â€"283).

2. **Breiman, L.** (2001). Random forests. *Machine Learning*, 45(1), 5â€"32.

3. **Friedman, J. H.** (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189â€"1232.

4. **Grinsztajn, L., Oyallon, E., & Varoquaux, G.** (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems (NeurIPS)*, 35, 507â€"520.

5. **Kadiwal, A.** (2020). *Water Quality Metrics for Potability Prediction*. Kaggle Dataset. Retrieved from https://www.kaggle.com/datasets/adityakadiwal/water-potability

6. **Pedregosa, F., et al.** (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825â€"2830.

7. **United Nations.** (2015). *Transforming our world: The 2030 Agenda for Sustainable Development (Goal 6)*. General Assembly resolution 70/1.

8. **World Health Organization.** (2017). *Guidelines for drinking-water quality: Fourth edition incorporating the first addendum*. World Health Organization. Geneva.